



Population Density Modelling



September 03 - 06, 2024.

Michael Mitchell, Casey Farren-Colloty, Matthew Fenlon

Problem Context

Ultimate Aim: We want to build a population density dynamical model for Ireland.

Why? With a population model that is geographically resolved, we can better anticipate where to place schools, hospitals, public transport, high-voltage power and water infrastructure, and so on. Minor improvements here can mean 10's or 100's of millions saved over the coming decade, with optimisations leading to a more stable and prosperous society.

Data setting: The geographical population data is provided at the level of electoral districts. This is given as a geojson file, although it may be useful to work with a graph representation.

Relevant Questions

Building a full population density model is a huge task. Here are some sub-questions that are worth answering, in increasing order of difficulty, with each one plausibly taken on by an individual in a group over the 3-4 days:

1. **Recruit Difficulty - Guaranteed Progress - Literature Scanning: What are the other relevant variables?** Population density as a lone variable cannot account for everything. We suspect other variables will feature strongly in a full model. **What is the relative importance of these variables?** Literature on this might make reference to [Principal Component Analysis](#), and [Correlation](#) analysis with the target (density). Also, where might the data for these variables be found? The following variables should give you a hint at the kinds of factors we imagine will feature (not exhaustive/ordered!):
 - a. Proximity to Schools
 - b. Proximity to Hospitals
 - c. House Prices
 - d. Housing planning applications
 - e. Major industry planning applications
 - f. Public Transport
 - g. Geographic Features
 - h. Social/Cultural Factors

2. **Recruit Difficulty - Very Likely Progress - Data Analysis: Is Ireland's population density Pareto/Power Law distributed?** In physics, some systems can admit scale invariant behaviour (see [this video](#) from 5 minutes - 20 minutes for the Ising model's scale-invariant phase transition). If Ireland's population density distribution follows a power law, this narrows the class of plausible model equations (e.g. [Maxwell's differential equations](#) for no charges/currents)! Tackling this question will involve:
 - a. Interpolating population data from an irregularly shaped graph, to a regular, finely grained grid.
 - b. Building a histogram of population densities.
 - c. Checking to see whether [Zipf's Law](#) / [Pareto Distribution](#) / [Gilbrat's Law](#) roughly fits the data.
 - d. Perform the fit, find the relevant distributional parameters.
 - e. (Harder) Investigate and report on what scale invariant equations of motion look like.

3. **Recruit Difficulty - Very Likely Progress - Algorithm Development: How will Ireland's Demographic Age profile evolve?** Aside from performing geospatial predictions, we need a good understanding of birth rates and death rates. One interesting option is probabilistic population forecasting:
<https://www.sciencedirect.com/science/article/pii/S0169207021001394>. Can you build such a model for Ireland's overall death rate, and total fertility rate? Equipped with this information, **can you devise a means to distribute these macroscopic, national population trends back onto the local population grid?**

4. **Regular Difficulty - Likely Progress - Algorithm Development: Can we get a simple simulation working?** We have provided the source code for a mapping of a simple differential equation, the [Diffusion Equation](#), onto a graph representation of our population data, and its evolution forward in time, so you can see what it looks like. **How do more sophisticated dynamical considerations enter this model?** For example,
 - a. Presently our graph-based model is distance-agnostic, including only neighbour information, with no notion of scale. How might scale be introduced to the model (can the [Distance Matrix](#) feature here), such that public infrastructure information features accurately into the model?
 - b. Is the pull of major population centres from rural areas faithfully represented as a non-local attractive potential?
 - c. How would such a non-local potential be dynamically evolved)?
 - d. How would a [stochastic component](#) be included in the model?
 - e. How would external emigration/immigration statistics enter into the model?

- f. If another relevant variable were included, such as school location, how would this information interface with the model?
5. **Regular Difficulty - Likely Progress - Methodology Development: How do we assess model stability/sensitivity to initial conditions?** With dynamical systems modelling, we must be vigilant to chaotic behaviour. [Chaotic behaviour](#) is understood to mean that minor differences in the initial configuration of a dynamical system will be blown up during the course of an evolution, resulting in system trajectories that look as though they were never close at any time in the past. This is informally known as the [butterfly effect](#). In our case, we know that the census data is not perfect. If, however, these imperfections manifest as enormous disparities in our model prediction, our model is basically useless. It is important that we can test this phenomenon in our case.
- How would we calculate the [Lyapunov exponent](#) of our system?** This number is a measure of how fast nearby initial conditions will diverge under the influence of a given differential equation. There is a closely related concept of a Lyapunov period, which captures the time for divergence, just as half-life captures time for a certain decay amount. This is intuitive for 1D systems/functions, but what does it look like/ how is it calculated here? This will give a rough precision-horizon of our model (but says nothing of its accuracy)!
 - How is a stability analysis performed on the model?** This assesses the sensitivity of the model's predictions to changes in the model's parameters.
 - The difficult thing in both instances is deducing the Lyapunov exponent/performing the stability analysis in a model with geospatially resolved output!
6. **Regular Difficulty - Likely Progress - Methodology Development: How do we assess model non-ergodicity?** An ergodic system is one in which a classical dynamical system, if given sufficient time, will visit every point in phase space which is energetically allowed. A system being ergodic is equivalent to saying that the system's time average equals the ensemble average of many such systems (Birkoff's theorem, see [here](#)). We should expect our system not to be ergodic; once population settles in a given region, the inertia of that newly settled population to stay put distinguishes it from the average of all other possible scenarios, breaking ergodicity. Properly articulating this inertial tendency is crucial for community engagement in policy simulation; we must strive to ensure communities understand we are building for them, not through them! **How do we assess the inertial tendency of population density?**
7. **Hardened Difficulty - Uncertain Progress - Advanced Data Analysis: Can we build an internal immigration map of Ireland?** We expect a simple diffusion equation on the
- 

graph not to work because the action is local: the flow of young people from Cavan to Dublin does not flow through Meath, they jump non-locally from Cavan directly to Dublin. There is a direct flow of people from major population centres to other major population centres: despite Dublin and Galway being geographically far away, for example, we expect that from an internal immigration standpoint, they are quite close together. **How might this closeness be represented mathematically, since it is not captured geographically?**

- a. Is it possible to create a [correlation matrix](#) between regions that relates the flow of people into certain regions with the flow of people out of others?
 - b. Is the data available sufficient to perform this task? Can the data be coarse-grained to improve statistical significance to make it work? Is there any other data sources that might better inform the creation of such a correlation matrix?
 - c. If the correlation matrix can be made, **can a [minimal spanning tree](#) be used to discover which regions are topologically close from an immigration standpoint?**
8. **Hardened Difficulty - Uncertain Progress - Advanced Algorithm Development: How might we build an error estimate for the prediction?** We need to accompany a prediction with some measure of error. Being able to say that there is an 80% chance that town A will grow faster than the other nearby town B is just as (if not more) important than giving a single figure forecasting the overall population growth in each case.
 - a. Multiscale error: **How does error improve as we lump multiple electoral districts together?** We should expect the statistical significance to improve; can you find how the error on a whole region is different to the error on the parts?
 - b. Bootstrapping: Assuming a stochastic (random) contribution to the model dynamics, **how do we faithfully represent the spread of repeated stochastic simulations geospatially?** This is straightforward for single variables, see the simulations of the Total Fertility Rate for France in: <https://www.sciencedirect.com/science/article/pii/S0169207021001394>. How will this carry over to our problem?
9. **Veteran Difficulty - Minor Progress is Impressive - Advanced Algorithm Development: How do we perform policy simulation with this model?** The ultimate goal is to better inform infrastructure building and societal decision making. How might this be investigated in our setting?
 - a. If there is an energetic description from which the dynamics of our system is derived, can a proposed policy decision (building a school here or there) be represented by a time-dependent potential?
 - b. If the system is evolved without a time-dependent policy scenario, can we devise a means to quantify a growing need for a school/hospital/public transport

infrastructure to be built? **What is the mathematical signature of a community's neglect?**

10. **Veteran Difficulty - Minor Progress is Impressive - Advanced Algorithm Development: Does Deep Learning have anything to offer this problem?** This problem will undoubtedly admit very non-linear behaviour, which may not be amenable to explicit mathematics within our capacity to hypothesise. Deep Learning could come to our rescue, but not without a very deliberate problem representation.
- a. Is there enough data to use Deep Learning without chronically overfitting? We only have 4-5 census snapshots, with ~3500 thousand electoral districts. Other independent variables risk exacerbating the difficulty of fitting the dynamics.
 - b. What architectures are best suited to such a problem? Convolutional graph neural networks are nice, but the convolution is a local operation, which we expect not to be super applicable to immigration dynamics. Reservoir computing is exceptional at handling chaotic and dynamical systems, but how might we manoeuvre the graph representation to plug into the usual reservoir computing input format? Transformer architectures have an attention mechanism which may work well here for non-local recontextualisation, but do we have the data to pull it off?
 - c. What are the societal risks associated with such a data-derived approach?

Acknowledgements

Matthew Fenlon, for cleaning the data, and making it coherent between census cycles.

School of Physics, for really pulling through in providing space.

School of Maths, for providing chairs and tables

Éirgrid, for contextualising the societal problem

Prof. Stefan Hutzler, for the insights from econophysics

Prof. Harun Siljack, for the insights from Complex systems

